

National University Of Computer & Emerging sciences (FAST)

(Karachi Campus)

Final Semester-Examination.2010

Course Code: MT104
Time allowed: 2h 45min

Linear Algebra
Fall 2010

Date: 28/12/2010
Max. Marks: 90

Instructions:

- 1) Attempt all questions. Show all necessary working and calculations.
- 2) Use blue or black marker to mention the question number on answer sheet.

Q1. Find the necessary and sufficient conditions for the existence of a solution to the following system. [08]

$$x + y + 2z = a_1$$

$$-2x - z = a_2$$

$$x + 3y + 5z = a_3$$

Q2. Solve the following system by Gauss Jordan elimination method. [10]

$$\begin{bmatrix} 0 & 0 & -2 & 0 & 7 & 12 \\ 2 & 4 & -10 & 6 & 12 & 28 \\ 2 & 4 & -5 & 6 & -5 & -1 \end{bmatrix}$$

Q3. Find the Eigen values and Orthonormal basis for the Eigen space of A where $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ [10]

Q4. Find Rank and nullity of the matrix $A = \begin{bmatrix} 1 & 4 & 5 & 2 \\ 2 & 1 & 3 & 0 \\ -1 & 3 & 2 & 2 \end{bmatrix}$ then verify that the values obtained

satisfy the dimension theorem for matrices. [10]

Q5. Find the standard matrix for the composition of transformations when a vector in R^3 is rotated counter clock wise about x-axis through an angle 60° , followed by a reflection in yz -plane, followed by a orthogonal projection on xz -plane. [08]

Q6. In each part use the given inner product on R^2 to find the $\|w\|$, where $w = (-1, 3)$. [08]

a) The weighted Euclidean product $\langle u, v \rangle = 3u_1v_1 + 2u_2v_2$ where $u = (u_1, u_2)$ and $v = (v_1, v_2)$.

b) The Inner product generated by the matrix $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$

Q7. Given $A = \begin{bmatrix} 2 & -1 & 1 \\ 4 & 1 & -3 \\ 2 & -1 & 3 \end{bmatrix}$, Evaluate determinant of A , and $\det(A^{-1})$. What is the relation between $\det(A)$ and $\det(A^{-1})$. [10]

Q8. a) Write the vector $V = (1, -2, 5)$ as a linear combination of the vectors $p = (1, -1, 1)$, $q = (1, 2, 3)$ and $r = (2, -1, 1)$ [08]

b) Determine whether the given vectors in R^3 are linearly independent or linearly dependent?
 $(1, -2, 3)$, $(5, 6, -1)$, $(3, 2, 1)$

Q9. Given $u_1 = (1, 1, 1)$, $u_2 = (-1, 1, 0)$ and $u_3 = (1, 2, 1)$ on R^3 have the Euclidean inner product. Apply Gram-Schmidt process to transform the basis $\{u_1, u_2, u_3\}$ into an orthonormal basis but at first Define Orthogonal and Orthonormal basis. [10]

Q10. V be the space spanned by $v_1 = \cos^2 x$, $v_2 = \sin^2 x$ and $v_3 = \cos 2x$. Show that $S = \{v_1, v_2, v_3\}$ is not a basis for V and then find a basis for V . [08]

Best of luck

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